Bayesian Networks (BN)

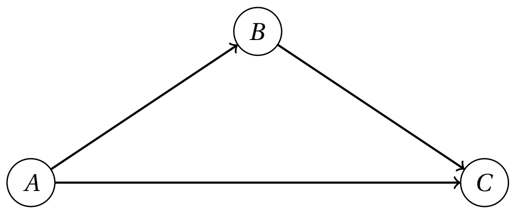
1 Definition

Definition 1.1 (Bayesian Network)**.**

A **Bayesian Network (BN)** is a probabilistic graphical model that represents a set of variables and their conditional dependencies.

- It is a **directed acyclic graph (DAG)**.
- Every **node** represents a **random variable** and is associated with a **conditional probability table (CPT)**.
- The **Local Markov Property** holds: each random variable is conditionally independent of its non-descendants given its parents.

An example



Conditional Probability Tables (CPTs)

A	$P(A)$
1	0.2
2	0.5
3	0.3

A	B	$P(B \mid A)$
1	T	0.7
1	F	0.3
2	T	0.4
2	F	0.6
3	T	0.9
3	F	0.1

A	1	1	1	1	2	2	2	2	3	3	3	3
B	T	T	F	F	T	T	F	F	T	T	F	F
C	$+1$	-1	$+1$	-1	$+1$	-1	$+1$	-1	$+1$	-1	$+1$	-1
$P(C \mid A,B)$	0.8	0.2	0.5	0.5	0.6	0.4	0.3	0.7	0.9	0.1	0.4	0.6

1.1 The CPTs

For a node without parents, the CPT simply gives the prior distribution of the variable.

For a random variable X_i with parents $\text{Parents}(X_i)$, the conditional probability table (CPT) specifies the probabilities $P(X_i \mid \text{Parents}(X_i))$.

To be more specific, let X be a node in the BN and let the set of its parents $\text{Parents}(X) = \{Y_1, Y_2, \ldots, Y_n\}$. Suppose:

- $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)$ is a tuple of the **parents** of X.
- $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$ denotes a variable that takes values of $\boldsymbol{Y}$.

The CPT of node X defines a multi-variable function as follows:

$$
\begin{aligned}
f(x, \boldsymbol{y}) &:= P(X = x \mid \boldsymbol{Y} = \boldsymbol{y}) \\
&= P(X = x \mid \{Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n\}) \\
&= P(X = x \mid \bigcup_{Y_j \in \text{Parents}(X)} \{Y_j = y_j\}).
\end{aligned}
$$

1.2 The Local Markov Property

Definition 1.2 (Conditional Independence)**.**

Let A, B, and C be three events. A is said to be **conditionally independent** of B given C, denoted as $(A \perp B) \mid C$, if and only if:

$$P(A \mid B \cup C) = P(A \mid C),$$

or equivalently,

$$P(A \cup B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

The Local Markov Property states that a node is conditionally independent of its non-descendants given its parents. Formally, for a node X in the BN:

$$X \perp \mathrm{NonDesc}(X) \ \mid \ \mathrm{Parents}(X),$$

or more accurately,

$$X \ \perp \ \mathrm{NonDesc}(X) \setminus \mathrm{Parents}(X) \ \mid \ \mathrm{Parents}(X).$$

This leads to the following equation:

$$P\big(X \mid \mathrm{Parents}(X) \cup \mathrm{NonDesc}(X)\big) = P\big(X \mid \mathrm{Parents}(X)\big)$$

2 Properties

Theorem 2.1 (Factorization Theorem / Chain Rule)**.**

Given a Bayesian Network with nodes $\boldsymbol{U} = \{X_1, X_2, \ldots, X_n\}$, the joint probability distribution of all random variables can be factorized as:

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) = \prod_{X_i \in \boldsymbol{U}} P\Big(X_i = x_i \mid \bigcup_{X_j \in \mathrm{Parents}(X_i)} \{X_j = x_j\}\Big)$$

Given the values $x_1, x_2, \ldots, x_n$, every factor on the RHS can be obtained from the CPT of a node.

***Proof*.**

The proof is based on three preliminary points:

1. General Chain Rule For any random variables $X_1, X_2, \ldots, X_n$, the joint probability distribution can be expressed as:

$$\begin{aligned} P(X_1, X_2, \ldots, X_n) &= P(X_1) \cdot P(X_2 \mid X_1) \cdot P(X_3 \mid \{X_1, X_2\}) \cdots P(X_n \mid \{X_1, X_2, \ldots, X_{n-1}\}) \\ &= \prod_{i=1}^{n} P\Big(X_i \mid \{X_1, X_2, \ldots, X_{i-1}\}\Big) \\ &= \prod_{i=1}^{n} P\Big(X_i \mid \bigcup_{j=1}^{i-1} \{X_j = x_j\}\Big) \end{aligned}$$

2. The Local Markov Property Let X be any node in the BN, $\mathrm{NonDesc}(X)$ be the set of its non-descendants, and $\mathrm{Parents}(X)$ be the set of its parents. $\mathrm{Parents}(X) \subseteq \mathrm{NonDesc}(X)$. The property gives:

$$P\big(X \mid \mathrm{NonDesc}(X)\big) = P\big(X \mid \mathrm{Parents}(X)\big)$$

3. Topological Ordering A DAG has at least one topological ordering, which means we can find an arrangement $X_1, X_2, \ldots, X_n$ such that for any X_i, $\mathrm{Parents}(X_i) \subseteq \{X_1, X_2, \ldots, X_{i-1}\}$.

Based on the above three points, let $X_1, X_2, \ldots, X_n$ be in topological order, then we have:

$$\begin{aligned} P(X_1, X_2, \ldots, X_n) &= \prod_{i=1}^{n} P\Big(X_i \mid \{X_1, X_2, \ldots, X_{i-1}\}\Big) \\ &= \prod_{i=1}^{n} P\Big(X_i \mid \text{NonDesc}(X_i)\Big) \\ &= \prod_{i=1}^{n} P\Big(X_i \mid \text{Parents}(X_i)\Big) \end{aligned}$$

□

3 Inference in a Bayesian Network

Some basic concepts:

- **Query**: The variables we want to know about their distributions.
- **Evidence**: The variables and their observed values.
- **Hidden Variables**: The variables that are neither query nor evidence.
- **Goal**: Compute the conditional probability distribution of the query variables given the evidence.

Different algorithms can be used for inference in Bayesian Networks, including:

- **Exact Inference**: Enumeration, Variable Elimination, Belief Propagation.
- **Approximate Inference**: Monte Carlo methods, Variational Inference.

3.1 Inference by Enumeration

- Query variable: X
- The tuple of evidence variables: $\boldsymbol{E} = (E_1, E_2, \ldots, E_m)$
- The tuple of observed values for evidence variables: $\boldsymbol{e} = (e_1, e_2, \ldots, e_m)$
- The tuple of hidden variables: $\boldsymbol{H} = (H_1, H_2, \ldots, H_k)$
- The tuple of values for hidden variables: $\boldsymbol{h} = (h_1, h_2, \ldots, h_k)$

$$\begin{aligned} P(X = x \mid \boldsymbol{E} = \boldsymbol{e}) &= \frac{P(X = x, \boldsymbol{E} = \boldsymbol{e})}{P(\boldsymbol{E} = \boldsymbol{e})} \\ &= \alpha \cdot P(X = x, \boldsymbol{E} = \boldsymbol{e}) \\ &= \alpha \cdot \sum_{\boldsymbol{h}} P(X = x, \boldsymbol{H} = \boldsymbol{h}, \boldsymbol{E} = \boldsymbol{e}) \\ &= \alpha \cdot \sum_{h_1} \cdots \sum_{h_k} P(X = x, H_1 = h_1, \ldots, H_k = h_k, \boldsymbol{E} = \boldsymbol{e}) \end{aligned}$$

where α is the normalization constant that can be found after computing $P(X = x, \boldsymbol{E} = \boldsymbol{e})$ for all values of x.
The term $P(X = x, \boldsymbol{H} = \boldsymbol{h}, \boldsymbol{E} = \boldsymbol{e})$ is a joint probability distribution and can be computed using the Factorization Theorem.

Time Complexity

Suppose:

- The BN has n nodes in total.
- Each random variable has at most d possible values.
- There are k hidden variables.

Further suppose reading from a CPT is $O(1)$, then computing each $P(X = x, \boldsymbol{H} = \boldsymbol{h}, \boldsymbol{E} = \boldsymbol{e})$ takes $O(n)$.

Conclusion The final time complexity is $O(n \cdot d^k)$, or $O(n \cdot d^n)$ as the evidence variables are usually only a few.

3.2 Variable Elimination

Concepts

1. Factors A **factor** is a multi-variable function that maps some values taken by a set of variables to a real number. For example, a factor $f(X,Y,Z)$ can be defined as:

$$f(x,y,z) = P(X = x \mid Y = y, Z = z)$$

CPTs are also factors.

The order of parameters of a factor does not matter. A variable is dependent on all other variables in the factors where it appears. The dependences among variables are not directed.

Scope of a factor The **scope** of a factor is the set of variables that the factor depends on. For example, the scope of factor $f(X,Y,Z)$ is $\{X,Y,Z\}$.

2. Pointwise product The **pointwise product** of two factors yields a new factor, whose scope is the union of the scopes of the two factors. Let $f_1(X_1, X_2, \dots, X_n, Z_1, Z_2, \dots, Z_k)$ and $f_2(Y_1, Y_2, \dots, Y_m, Z_1, Z_2, \dots, Z_k)$ be two factors with common variables $Z_1, Z_2, \dots, Z_k$. The pointwise product $f = f_1 \cdot f_2$ is defined as:

$$f(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m, z_1, z_2, \dots, z_k) := f_1(x_1, x_2, \dots, x_n, z_1, z_2, \dots, z_k) \cdot f_2(y_1, y_2, \dots, y_m, z_1, z_2, \dots, z_k)$$

3. Summing out Factors show the dependences among variables. When a variable only appears in one factor, we can conclude that it only depends on the variables in that factor. Thus, we can **sum up** all possible values of that variable to eliminate it from the factor.

Let $f(X, Y_1, Y_2, \dots, Y_n)$ be a factor. The operation of **summing out** variable X from factor f yields a new factor ϕ defined as:

$$\phi(y_1, y_2, \dots, y_n) := \sum_x f(x, y_1, y_2, \dots, y_n)$$

The Algorithm

Throughout the algorithm, we maintain a set of factors.

1. The initial set of factors The initial factors are the CPTs of all nodes in the BN.

Evidence Instantiation If a factor contains evidence variables, we restrict the factor to the observed values. For example, $f(x,y,e) := P(X = x \mid Y = y, E = e)$, and we have observed $E = e^*$, then we replace the factor f with $f'(x,y) := f(x,y,e^*)$.

2. Pointwise product and summing out Pick a hidden variable H to eliminate.

1. Identify all factors $f_1, f_2, \dots, f_k$ that contain variable H.
2. Compute the pointwise product of these factors to get a new factor f.
3. Sum out variable H from factor f to get a new factor ϕ that does not contain H.
4. Add factor ϕ to the set of factors and remove $f_1, f_2, \dots, f_k$.

3. Repeat Repeat the above step until a factor containing only the query variable X is obtained and no other factors contain X.

4. Normalization The remaining factor $\tau(X)$ gives an unnormalized distribution over X. Normalize it to get the final result:

$$P(X = x \mid \boldsymbol{E} = \boldsymbol{e}) = \frac{\tau(x)}{\sum_{x'} \tau(x')}$$

Time Complexity

Suppose:

- The BN has n nodes in total.
- Each random variable has at most d possible values.

A factor involving m variables has d^m entries.

The pointwise product of two factors with m_1 and m_2 variables results in a new factor with at most $m_1 + m_2$ variables, and takes $O(d^{m_1+m_2})$ time to compute.

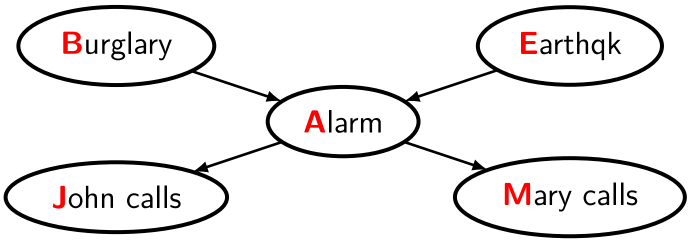
Summing out a variable from a factor with m variables results in a new factor with $m-1$ variables, and takes $O(d^m)$ time to compute.

The bottleneck occurs when doing pointwise products or summing out.

Conclusion The time complexity is $O(n \cdot d^{w+1})$, where w is the treewidth of the graph. In the worst case, w can be about n.

Another View of the Algorithm

We use an example to illustrate the process of variable elimination, to provide another perspective to understand the algorithm.



Given observations that $J = j$ and $M = m$, we want to compute the probability of burglary B. We start with the formula in the enumeration method:

$$
\begin{aligned}
P(B \mid j, m) &= \alpha \cdot \sum_e \sum_a P(B, e, a, j, m) \\
&= \alpha \cdot \sum_e \sum_a P(B) \cdot P(e) \cdot P(a \mid B, e) \cdot P(j \mid a) \cdot P(m \mid a) \qquad \text{by the Chain Rule} \\
&= \alpha \cdot \sum_e P(e) \cdot \sum_a P(a \mid B, e) \cdot P(j \mid a) \cdot P(m \mid a) \\
&\qquad\qquad P(B),\ P(e),\ P(a \mid B, e),\ P(j \mid a),\ P(m \mid a) \text{ correspond to 5 factors.} \\
&= \alpha \cdot P(B) \sum_e P(e) \cdot \sum_a f_1(B, e, a) \\
&\qquad \text{3 factors are merged. This is equivalent to performing pointwise products.} \\
&= \alpha \cdot P(B) \sum_e P(e) \cdot \phi_1(B, e) \qquad \text{Variable } A \text{ is summed out.}
\end{aligned}
$$

In the enumeration method, we do not factor out the common factors. In variable elimination, we extract common factors to reduce redundant computations.

But different order of eliminating variables leads to different sizes of intermediate factors, which may lead to different time complexities when performing pointwise products and summing out.

$$
\begin{aligned}
P(B \mid j, m) &= \alpha \cdot \sum_e \sum_a P(B) \cdot P(e) \cdot P(a \mid B, e) \cdot P(j \mid a) \cdot P(m \mid a) \\
&= \alpha \cdot P(B) \sum_a P(j \mid a) \cdot P(m \mid a) \sum_e P(e) \cdot P(a \mid B, e) \quad \text{The summation order can be switched.}
\end{aligned}
$$

Finding the optimal order of eliminating variables is NP-hard. But there are some heuristics to find a good order that usually works well in practice. For example, the **min-fill** heuristic chooses the variable that minimizes the number of edges added to the graph when the variable is eliminated.